

Chapter 6: Digital Terrain Modelling

1. Introduction

In traditional cartography, three-dimensional land surfaces were awkwardly represented on two-dimensional paper sheets using abstract static lines (contours) or color tints. Digital Terrain Modelling (DTM) modernizes this process by using computational data structures to continuously map the Earth's topography. This allows engineers and geoscientists to dynamically measure, simulate, and visualize elevations, slopes, and landscape shapes mathematically.

2. Definitions and Terminology

To avoid structural confusion in geospatial engineering, specific distinctions must be made between key topographic modeling terms:

- **Digital Elevation Model (DEM):** A generic or specific digital representation of continuous elevation values over a topographic surface. Crucially, a bare-earth DEM represents the bare ground surface, stripping away temporary surface objects like buildings, infrastructure, and forest canopies.
- **Digital Terrain Model (DTM):** In strict academic engineering terms, a DTM is a bare-earth DEM that has been augmented with structural linear landscape features such as **breaklines** (e.g., ridge centerlines, stream beds, cliff edges) to capture abrupt changes in topographic smooth slopes.
- **Digital Surface Model (DSM):** A digital representation of the earth's surface that includes the absolute top elevations of all natural and man-made features (e.g., tree crowns, building roofs, power lines) resting on the landscape.

Core Data Representation Structures

- **Regular Grid (Raster DEM):** Space is divided into a continuous matrix of square cells, where each pixel coordinate contains a single Z -value representing average elevation.

It is simple to compute but struggles to accurately model jagged, high-relief micro-topography without exponentially expanding file sizes.

- **Triangulated Irregular Network (TIN):** A vector-based topological data structure that manages irregular, non-overlapping triangles formed by connecting coordinate data points (X, Y, Z) via Delaunay triangulation.
 - It is highly efficient because it stores dense clusters of points only where the topography changes drastically (e.g., mountain peaks, valleys) and uses fewer points across flat plains.

3. Acquisition of Digital Terrain Models

Elevations used to construct digital models are captured through primary field tracking or secondary remote sensing methodologies:

A. Ground-Based Methods (High Accuracy, Limited Extent)

- **Direct Field Surveying:** Collecting precise (X, Y, Z) point configurations on site using high-end Total Stations, digital levels, and GNSS/GPS receivers. This delivers sub-centimeter structural accuracy but is highly labor-intensive and limited to small localized project boundaries.

B. Remote Sensing & Airborne Methods (Massive Extent, High Density)

- **Photogrammetry:** Processing overlapping pairs of aerial or satellite photographs (stereoscopic imagery). By matching pixels from two different viewing perspectives, geometric parallax calculations compute explicit surface elevations.
- **Airborne LiDAR (Light Detection and Ranging):** An active remote sensing technique where an aircraft shoots millions of rapid laser pulses down to the ground. By measuring the precise time delay of the laser echo return, it generates a high-density 3D coordinate **point cloud**, which can filter out canopy leaves to map both the absolute treetop (DSM) and the bare earth ground (DTM) below.

- **Radar Interferometry (InSAR):** Utilizing phase differences between bouncing radar wave signals captured by satellite sensors (such as the historic Shuttle Radar Topography Mission - SRTM) to map massive, continental-scale elevation matrices regardless of cloud cover.

4. Data Processing

Raw captured topographic data points must undergo processing to turn disjointed measurements into clean, usable surfaces:

- **Filtering and Classification:** Especially critical for LiDAR datasets. Algorithms separate the raw point cloud data into ground returns (bare soil, rock) and non-ground returns (vegetation cover, buildings) to allow for the separate construction of pure DTMs versus DSMs.
- **Spatial Interpolation:** Because it is impossible to sample the elevation of every single square inch of the Earth, computers apply spatial interpolation math to estimate unknown heights between known sample data points. Common techniques include:
 - **Inverse Distance Weighting (IDW):** A deterministic approach assuming that closer sample points have a greater mathematical influence on the unknown cell target value than points further away.
 - **Kriging:** A highly advanced geostatistical interpolation method that evaluates both the distance and the structural spatial orientation trends among surrounding points to provide optimal, statistically unbiased estimates.
- **Void Filling and Artifact Removal:** Identifying and cleaning up systemic sensor dropouts, random data holes, or artificial spikes (errors) to create a smooth, hydraulically continuous surface profile.

5. Analysis and Visualization

Once a terrain model is constructed, a GIS can dynamically calculate derivative landform layers for advanced structural analysis:

A. Primary Analytical Derivatives

- **Slope Gradient:** Measures the maximum rate of change in elevation values from a central cell to its immediate neighbors, expressed in degrees $(0^\circ \text{ to } 90^\circ)$ or percent slope.
- **Aspect (Direction):** Calculates the compass orientation direction that a sloping hill face is pointing (e.g., North, Southeast), measured clockwise from 0° (North) to 360° . This is critical for solar radiation modeling and agricultural micro-climate evaluation.
- **Curvature:** Measures the structural shape of a slope—whether it is convex (pointing outward, accelerating water runoff) or concave (dishing inward, accumulating water flow).

B. Terrain Visualization Techniques

- **Hillshading (Shaded Relief):** Simulates the localized lighting effects of an artificial sun positioned at a specific angle and altitude in the sky. This casts realistic highlights and shadows across a 2D map, making it visually intuitive to see ridges and valley drop-offs.
- **Perspective 3D Views:** Projecting the grid matrix or TIN wireframe mesh into three-dimensional block diagrams, allowing engineers to rotate, fly through, or view terrain designs from a realistic ground perspective.

6. Applications of Digital Terrain Models

Digital Terrain Models serve as core decision-making frameworks across many technical fields:

- **Hydrological Modeling and Watershed Management:** DTMs allow software to track how water flows across a landscape. By analyzing downhill slopes, models can automatically map out drainage pathways, stream networks, watershed boundaries, and delineate flash-flood hazard zones.
- **Civil Engineering and Infrastructure Design:** Crucial for planning highways, railways, and pipelines. Engineers use DTM profiles to perform **cut-and-fill calculations**,

balancing how much dirt must be dug out versus how much must be piled up to maintain a flat, stable engineering grade.

- **Line-of-Sight and Visibility (Viewshed) Analysis:** Calculating whether a direct line of sight exists between an observer point and a target landscape. This is vital for telecommunication companies selecting optimal sites for cell towers, or military planners map-analyzing radar blindspots.
- **Natural Hazard Mitigation:** Used to monitor landslide risks, model avalanche paths, track mudflow directions, and predict coastal storm surge inundation levels based on absolute elevation buffers.